

On replacing humans with human simulators in human-in-the-loop systems built to interact with humans[☆]

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Abstract

With the rise of large language models (LLMs), which can generate fluent text and follow users’ instructions, many attempts have emerged to replace humans on a variety of occasions: data annotation, customer service, online tutorials, and even user testing. While it might seem that LLMs could easily take over many of these tasks—especially those that are already highly automated with minimal human involvement—we want to take a step back and consider whether these promising replacements would truly respond to the reasons we put humans in the loop of computer systems in the first place. In this chapter, we present a case study focusing on a specific area of research, namely interactive crowd-powered systems. These systems incorporate human crowd workers, who can be recruited and managed rapidly and programmatically from the internet, into the loop of computer systems to interact with users in (nearly) real-time. We first overview the history of such systems, discuss the original motivations for having human workers within these systems, and look into a series of recent attempts to replace these humans with LLMs. Realizing some gaps within these efforts, we then zero in on an interesting case study, comparing two projects in parallel: both studied users’ interactions with voice-based smart speakers that can hold open-domain conversations, one powered by an LLM, and the other powered by human workers. Through this comparison, we argue that while last-mile interaction issues such as conversation cut-offs and persistent speech recognition problems across both human-powered and LLM workflows, each approach requires distinct considerations. Specifically, certain accommodations must be made when replacing human workflows with LLMs.

[☆]This chapter was developed and extended based on our short piece published at AAAI Symposium Series 2024 (Huang et al., 2024).

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LLMs may introduce new challenges that are rarely encountered by human workers, and vice versa.

Keywords

Case study, Large language models, Data annotation, Customer service, Human crowd-powered systems

11.1 Real-time crowd-powered systems & why we built them

While crowdsourcing, as well as crowd-powered systems, may seem like modern concepts primarily used for internet-based activities, they have been part of society for a long time and have been applied in various contexts.

For example, it was not until the mid-19th century that modern weather forecasting, as we know it today, began to take shape with the invention of the electric telegraph. Joseph Henry, the first secretary of the Smithsonian, initiated the weather observation project that utilized the telegraph to collect weather descriptions around the country (Smithsonian Institution Archives). These observations were then displayed on maps to depict daily weather conditions, introducing the earliest form of weather forecasting. The project was later transferred to the US government and eventually turned into the National Weather Service. It is also noteworthy that the amount of weather observations received by the institute was too large to retain and created a massive backlog of data and the human “computers” who were employed to process the data.

In more recent applications, crowdsourcing has been leveraged to tackle tasks that are too large for a small group of researchers to handle. Galaxy Zoo was able to classify a large number of galaxies with the help of ordinary people online (Lintott et al., 2008); with the public’s assistance, FoldIt gamified the process of protein folding, which further helped the research of protein structure prediction (Cooper et al., 2010); one inspiration for reCAPTCHA was to digitalize books and documents that were not legible for computers via scanning, therefore, with the help of the crowd, it tried to identify the characters in a scan to help its digitization (Von Ahn et al., 2008).

While many tasks in the past have engaged the crowd to help solve the problem at hand, it was not always in a timely fashion and may have required a long period of time to solve the problem. Also previous crowdsourcing projects were of a larger scale and not necessarily helping users solve their personal problems. In

the past decade, with the advent of platforms like Amazon's Mechanical Turk and the widespread use of personal mobile devices such as smartphones, the concept of crowdsourcing was introduced to users on a more personal level; it has also been used to help users with their needs in (nearly) real-time fashion, such as Evorus (Huang et al., 2018) and VizWiz (Bigham et al., 2010). Evorus is a crowd-powered conversational assistant that is capable of responding to users' inquiries in a timely fashion. VizWiz, on the other hand, is able to quickly respond to visual questions sent by blind users or those with visual impairments.

Taking a holistic look at the development of crowdsourcing over the years, besides intellectually motivated explorations (e.g., can this be done this way?), there are two main aspects crowdsourcing has been aiming to solve: (i) closing the social-technical gap for **humans**, and (ii) improving **computer** technologies.

For a long time, the argument for (nearly) real-time crowd-powered systems—employing online workers to operate computer components for quick judgments or predictions, mimicking actual computer components—was to **fill the social-technical gaps of current automatic technologies, enabling researchers to study interaction problems or user needs that can only be studied when particular systems or services exist**. For example, Huang et al. suggested in Evorus (Huang et al., 2018) a crowd-powered conversational assistant:

“Such automated [general conversational assistant] systems only become useful once they can completely take over from the crowd-powered system. Such abrupt transition points mean substantial upfront costs must be paid for collecting training examples before any automation can be tested in an online system. We explore an alternative approach of a crowd-powered system architecture that supports gradual automation over time.”

VizWiz (Bigham et al., 2010) also said:

“Talking mobile devices from both research and industry have been designed to help blind people solve visual problems in their everyday lives, but current automatic approaches are not yet up to the task. Products designed for blind people are specialized for a few functions, are prone to errors, and are usually quite expensive.”

Soylent (Bernstein et al., 2010) is another example:

“The main contribution of this paper is the idea of embedding paid crowd workers in an interactive user interface to support complex cognition and manipulation tasks on demand. These crowd workers do tasks that computers cannot reliably do automatically and the user cannot easily script.”

This was essentially based on the stance: “AI is not there (yet); let’s use humans for now”. A classic example is VizWiz (Bigham et al., 2010), a nearly real-time crowd-powered system that uses online crowd workers to quickly respond to visual questions sent from blind users or those with visual impairments. Well before AI could reliably answer arbitrary visual questions, VizWiz enabled large-scale research into the visual challenges that blind users encounter daily. Other impactful examples include Chorus (Lasecki et al., 2013), Soylent (Bernstein et al., 2010), and Scribe (Lasecki et al., 2012). A series of techniques and tools were created to make these systems possible, such as the retainer model (Bernstein et al., 2011) and quikturkit (Bigham et al., 2010), or even scalable (Huang and Bigham, 2017). Some projects, such as Evorus (Huang et al., 2018), a follow-up framework that aimed to automate Chorus (Lasecki et al., 2013) over time, and VizWiz Challenge (Gurari et al., 2018), a computer-vision shared task that aimed to build automated question-answering systems for questions collected via VizWiz, took the process a step further to argue that the insights gained from studying these imminent problems could, in turn, improve today’s computer technology.

11.2 Attempting to replace humans in human-in-the-loop systems with LLMs and what could possibly go wrong

The key argument that motivates crowd-powered systems—that AIs are not yet capable enough, and, therefore, humans must be involved to bridge socio-technical gaps—has been implicitly challenged by the rise of LLMs. ChatGPT was introduced to the public in November 2022, and since then, numerous LLMs have been developed and made widely accessible, including Llama by Meta (Touvron et al., 2023) and Gemini by Google (Gemini Team et al., 2023). These models can generate fluent responses and handle open-ended user requests, with some demonstrating reasoning abilities. As a result, people have increasingly turned to LLMs to replace humans in various roles—justifiably or not—such as replacing customer service agents with chatbots (Klarna, 2024; Amazon; IBM), reducing the need for writers or editors by employing AI-based editors (Germain, 2024; De Vynck, 2024), and even introducing conversational AI in mental health services (Filenko et al., 2024; Cheng et al., 2023; ChatBeacon; Koko). One of the more extreme examples involves replacing human participants with LLMs in psychological surveys or behavioral studies (Dillion et al., 2023).

In the realm of crowdsourcing, studies have shown that ChatGPT outperforms online crowd workers in tasks such as labeling and writing (Törnberg, 2023; Gilardi et al., 2023; Cegin et al., 2023; He et al., 2024). Many crowd workers now secretly rely on ChatGPT to assist with their tasks—a study found that even 20 aggregated crowdsourced labels from a well-executed MTurk pipeline were less accurate than GPT-4’s labels (He et al., 2024). These findings collectively suggest that LLMs may be closing the socio-technical gaps that once necessitated real-time crowd-powered systems more effectively than human crowd workers. If this trend continues, researchers might opt to replace crowd workers with LLMs in these systems, which could significantly reduce both engineering efforts and costs while still enabling the study of the same set of interactive problems and user needs. Some studies have directly compared crowdsourcing workflows with LLM workflows, aiming to understand which aspects of crowdsourcing can be adapted to LLM workflows and which cannot (Wu et al., 2023; Grunde-McLaughlin et al., 2023).

Looping back to the two goals people wanted to achieve when they put crowd workers inside crowd-powered systems (i.e., helping human users and improving computer technologies), one interesting question is: are these LLM-powered components truly effective in meeting these goals? More specifically, as we replace humans in human-in-the-loop systems with LLMs—which is already happening or being considered in many areas—are we losing any capabilities that human workers still perform better than LLMs?

11.3 A tale of two systems

With the aim of addressing this question, we were given an interesting opportunity to compare two systems, which we present in this section. Both systems are built on Amazon’s Echo, a commercial smart speaker integrated with Amazon Alexa. One system, developed by us (the authors of this chapter), incorporated humans-in-the-loop to facilitate conversations (Huang et al., 2022), while the other, created by a team of researchers from Johns Hopkins University and Northeastern University, utilized an LLM to manage conversations behind the scenes (Mahmood et al., 2023).

In this case study, we focus on the potential impact of replacing humans, particularly within real-time, crowd-powered interactive systems that are designed to respond quickly enough to maintain synchronous interactions with users. Due to the interactive nature of these systems, they often rely on human workers behind the scenes to provide timely, high-quality responses—a task that aligns well with the capabilities of LLMs.

11.3.1 ECHOPAL: a human-in-the-loop voice assistant

ECHOPAL is a human-in-the-loop voice assistant based on Amazon Echo introduced by Huang et al. in 2022 (Huang et al., 2022). When a user talks to ECHOPAL, Echo records the audio and turns the speech into text through the built-in automatic speech recognition (ASR) system. The transcribed text is then sent to the backend of ECHOPAL where it is presented to the human worker as a message in the worker interface, and where the worker can see the transcribed message along with a set of possible alternative transcriptions generated by the system to address potential ASR errors. Furthermore, ECHOPAL also automatically generates and presents a list of suggested responses to the worker. As each alternative transcription has its own suggested responses, the worker can click on the transcription to switch between them. Search support sites, such as Google Search or Google Weather, are also provided in the interface. With the support of these technological features, the worker was required to produce each response within 25 seconds. A worker's response is then sent back to the Echo device, where a built-in text-to-speech (TTS) system reads out the message to the user. Users were allowed to chat freely with ECHOPAL without constraints on the conversation topic.

ECHOPAL inherited the spirit of Chorus and Evorus, which aimed to use human intelligence to fill the gaps between what users want to use conversational assistants for and what conversational assistants are capable of, allowing researchers to study interaction questions that could not otherwise be studied (Lasecki et al., 2013; Huang et al., 2018). In the user study, Huang et al. used participants as both users and workers (instead of hiring online workers to be the workers.)

11.3.2 CHATGPT-IN-ECHO: an LLM-in-the-loop voice assistant

CHATGPT-IN-ECHO was presented by Mahmood et al. in 2023, with the aim of exploring the constraints and opportunities of LLM-powered voice assistants (Mahmood et al., 2023).¹ In the same period of time, many other projects also appeared to connect LLMs, especially ChatGPT provided by OpenAI via an API² call, with voice assistants such as Alexa or Google Assistant (Yang et al., 2024; AndroidAuthority, 2023).

¹Mahmood et al. did not name their system. We used CHATGPT-IN-ECHO as a placeholder name for communication purposes.

²API (Application Programming Interface) is a set of tools that allow different programs to communicate with each other even if they are built differently.

When a user talks to CHATGPT-IN-ECHO, Echo records the audio and turns the speech into text through the built-in automatic speech recognition (ASR) system. The transcribed text is then sent to a primary middleman API; this middleman API is used to handle the time-out constraint set by Alexa Skill. If it does not receive a message from ChatGPT within a certain amount of time, it will initiate filler/small talk to maintain Alexa Skill activity. A secondary middleman API is used to communicate with ChatGPT. Upon receiving the relayed user transcription from the first middleman API, it will request ChatGPT via API call with the latest user message. A shared database of conversation history and progress indicated that communication were archived between the primary and secondary middleman API. Once the secondary middleman API receives the ChatGPT response, it will update the conversation history. The primary middleman API will monitor the database for conversation updates and send the ChatGPT message to the user through Alexa TTS. Three tasks from different scenarios were tested: medical self-diagnosis, creative planning, and discussion with opposing stances.

11.3.3 What makes the two comparable (or not)

Device: Amazon's Echo. ECHOPAL and CHATGPT-IN-ECHO were both run on Amazon's Echo devices, a commercial voice-enabled smart speaker designed to assist users via voice. In the original study, ECHOPAL ran on Echo (2nd generation), and CHATGPT-IN-ECHO ran on Echo Dot. Both devices had no monitors and used voice as the only communication channel with users, aided with a ring light to show the state of the device, e.g., listening, processing, or deactivated.

Note that Alexa Skill did not (and still does not) support audio streaming and only provided the transcribed text of the users' speech to the backend for developers. The transcribed texts were fed directly to the backend without utilizing the intent architecture of Alexa Skill. The conversation logs between the user and the voice assistants were stored in the database hosted outside of the Alexa Skill.

Schematics. ECHOPAL and CHATGPT-IN-ECHO are similar also in terms of their schematics. Both utilized a behind-the-scenes architecture to handle the conversation history, and Alexa Skills interacted with the backend responder regarding the user message. ECHOPAL had human workers—hence the need for a worker interface—and helper functions to help the workers. CHATGPT-IN-ECHO utilized an LLM and therefore needed to handle the API call.

Alexa API Time-Out. To our knowledge, the current default time limit for Alexa Skills to process responses is about 8 seconds, which is fixed for most cases. CHATGPT-IN-ECHO employed middleman APIs to handle the time-outs by inserting fillers and small talk before the 8-second time-outs had occurred and continuing the conversation after responses were received from ChatGPT. ECHOPAL, on the other hand, does not face the same issue because it was developed under an earlier version of Alexa Skill, and extending the time-out value to 25 seconds was possible. However, we believe that it is possible to modify ECHOPAL's architecture to fit the current Alexa Skill configuration and adopt techniques used by CHATGPT-IN-ECHO.

We (the authors of this chapter) are the authors of ECHOPAL (Huang et al., 2022), and the following comparison was made by us reading through the paper of CHATGPT-IN-ECHO in detail (Mahmood et al., 2023). We also looked into our experimental records to compare the details, which might include some nuanced information that we did not catch or mention in our original paper.

In the following, when we refer to “human,” we are specifically referring to the crowd workers involved in the operation of the voice assistant, not the users interacting with it.

11.3.4 Human problems that LLMs do not have

Long Response Latency. One of the main drawbacks for ECHOPAL is the long latency for its users, when they have to wait for an average of 17.68 seconds to get quality responses from Alexa. However, the same response time was considered too short for the workers at the backend to provide quality responses. Most of this latency comes from human workers needing to type; some latency comes from ASR and internet transmission time. Meanwhile, CHATGPT-IN-ECHO is powered by GPT-3.5, which can produce thousands of words in seconds.

To some extent, integration with LLMs might make future voice assistants that aim to hold longer, open-ended conversations one step closer to reality: it reduces the unreasonably long latency of human-powered voice assistants, it makes such systems more scalable and affordable to build than recruiting human workers', and it bypasses the speed limits set by humans typing scripts.

11.3.5 LLMs' problems that humans do not have

Oversharing and Repetitiveness. CHATGPT-IN-ECHO raised concerns regarding voice assistants providing repetitive information and oversharing, which was not observed in ECHOPAL's study.

Repetitive information provided by ChatGPT was reported in the scenarios tested by CHATGPT-IN-ECHO. Since voice users cannot review previous conversations unless recorded compared to text conversations, repetitive information causes a further inconvenience during the conversation. This phenomenon was particularly true in medical-related conversations; repetitive information can occur even if ChatGPT was prompted not to repeat certain warning information. OpenAI's policy likely causes this, and it might not be easily fixed by outside developers. Additionally, CHATGPT-IN-ECHO encountered a challenge with ChatGPT's over-sharing, where the responses provided by ChatGPT were too overwhelming for users to absorb. Even when prompted to give brief responses under 100 words, the higher information density in its text interactions, which ChatGPT is based on, can quickly overwhelm users in a voice setup.

11.3.6 Problems that both LLMs and humans have

Cut-Offs. We define cut-offs such as is Alexa stopping listening to a user's speech while a user is in the middle of its speech or intends to keep speaking. In ECHOPAL, cut-offs were attributed to (1) users having longer pauses between words, causing Alexa to think the user's speech had ended; and (2) Alexa entered listening mode after its own speech and stopped listening if users did not speak in a given time. CHATGPT-IN-ECHO faced similar problems, such as (1) partial listening, where partial speech was captured in a user's speech; (2) interruption, where Alexa interrupts the user's speech mid-sentence; and (3) pauses between words when the users organized their thoughts.

ASR Errors. ASR errors were one of the main obstacles in ECHOPAL, occurring when the audio of a user's speech was not obtainable and only the transcribed text by Alexa was received. Different alternative transcriptions were provided to help human workers to tackle this problem. CHATGPT-IN-ECHO also faced ASR problems but had and needed the inherent ChatGPT recovery mechanism, such as apologizing or asking the user to clarify. We suspect the typo fixing of the text-based system, which ChatGPT is good at, does not work as well for voice-based communications. This is likely because similar sounds do not translate to good guesses by ChatGPT.

Conversation Breakdowns. Both Cut-offs and ASR errors can cause the interaction to break down, meaning the current interaction session will either be (1) terminated by Alexa without the users' acknowledgment, or (2) require initiating a recovery mechanism in order to continue the interaction. The only solution to

address the first type of breakdown is for the user to reinitiate the Alexa Skill, effectively restarting the interaction. One way to make the reinitiation easier is by better maintaining the previous interaction history, but it still requires the user to actively restart the interaction. For the second scenario, recovery within the same interaction session is possible. Recovery can be initiated by the users or by the voice assistants. In the first category of a breakdown, users will need to actively restart the interaction with the voice assistants. In the second category, the voice assistants can assist in the recovery process within the same session. However, most recovery happens when users repeat the previous sentence that failed to be correctly processed, or they move on to the next sentence and discard or ignore the failed sentence.

One feature of echo devices is the light ring that indicates the state that Alexa is in. When the device is in “listening” mode, the light ring will illuminate and shimmer. ECHOPAL reported that users’ familiarity level with Echo devices’ interaction pattern significantly impacted their ability to handle cut-offs. Those who are more familiar with the Echo interaction patterns encountered fewer cut-offs compared to those with little or no familiarity.

CHATGPT-IN-ECHO described users’ mental model while interacting with voice assistants in multiturn taking scenarios. They suggested that actions users take after conversation breakdowns indicated the gap between users’ mental models and voice assistants’ capabilities. (The light ring on the device was not discussed in CHATGPT-IN-ECHO’s paper.)

Long responses are not suitable. ECHOPAL suggested that long text responses are not suitable for voice application since it takes a lot of time to read out the full paragraph and interrupt the flow of a voice conversation. CHATGPT-IN-ECHO, on the other hand, limited the response of ChatGPT to 100 words in prompts but still faced the problem of voice assistant oversharing and overwhelming the users.

11.4 When replacing humans with LLMs in interactive systems

The comparison between the two systems guided our response to the question of whether replacing humans with LLMs can effectively address the goals that lead us to include humans in interactive systems. LLMs can indeed fulfill *some* of these goals, particularly by reducing response latency—likely a key factor in the success of interactive systems—and by providing significantly better scalability than purely human-operated systems could achieve.

However, we also believe such replacements must be approached with caution, as humans are still far superior in handling the nuances of open-ended conversations and are more resilient in managing outlier situations, such as interpreting jokes in dialogue. In this section, we expand on our findings and offer several design implications.

11.4.1 Time vs. quality trade-offs in human-operated systems

We identified a trade-off between the quality of responses provided by ECHOPAL and the time required to generate them. In order to deliver higher-quality responses, workers needed more time. Participants assigned as workers operating ECHOPAL suggested that the 25-second response time was not enough. Enforcing a shorter response time would likely lead to a decrease in response quality. Grunde-McLaughlin et al. proposed the design space for LLM chaining by adapting crowdsourcing workflows (Grunde-McLaughlin et al., 2023). The trade-off between time and quality we observed in ECHOPAL also exists within the crowdsourcing aspect of this design space.

In contrast, for LLM-based systems such as CHATGPT-IN-ECHO, which utilized API calls instead of waiting for human workers to provide their responses, we speculate that such trade-offs to be less prevalent. While the latest OpenAI o1 model (OpenAI, 2024) emphasizes on its ability to “think” before replying to the users’ requests, causing it to respond more slowly. It is unclear to us whether similar trade-offs can be observed in LLM-based systems as newer models are introduced.

11.4.2 Effort needed in prompt engineering vs. microtask design

While an LLM’s workflow can offer lower latency and cost compared to crowdsourcing workflows, efforts must be directed toward prompting LLMs to generate more diverse responses. Due to the nature of crowdsourcing workflows, which incorporate multiple individual workers, it is possible to receive diverse responses by querying different workers with the same requests (Bigham et al., 2010; Huang et al., 2020; Grunde-McLaughlin et al., 2023). LLMs, on the other hand, need to rely on methods such as querying different models, adjusting model parameters and modifying prompts to produce diverse responses (Grunde-McLaughlin et al., 2023). However, elements of diversity inherent to real human are not well portrayed by LLMs, such as portraying various identity groups

(Wang et al., 2024) and representing pluralistic human values and perspectives (Sorensen et al., 2024). Additionally, instructions and tasks suitable for crowdsourcing workflows may be overly complicated for LLMs (Grunde-McLaughlin et al., 2023). In such cases, tasks need to be broken down into simpler and more specific components than the microtasks designed for human workers to allow LLMs to solve them effectively. Therefore, simpler instructions and tasks are likely required to utilize LLM capabilities effectively.

A more fundamental question arises: which requires more effort—designing a good workflow for LLMs or creating a human-computation workflow and interfaces for crowd workers? While it is tempting to conclude that designing crowdsourced microtasks is more time-consuming—since it requires a human-readable interface and extensive testing, which can be costly and time-intensive—we are hesitant to make such a claim. Human-computation workflows, though expensive to develop and test, can capitalize on assumptions about human workers, such as their diverse backgrounds, common sense knowledge, and understanding of social norms and dialogue, which can, in turn, reduce the effort needed to get things right. A more comprehensive comparison between the two approaches, however, would require significantly more research to determine.

11.4.3 Design implication

Based on the comparison we just presented and our experience with crowd-powered systems, two design implications have emerged.

11.4.3.1 Switching between infrastructures designed for humans and automated components presents significant design challenges due to their differing assumptions

Switching between infrastructures designed for human input and automated components presents significant design challenges due to their differing underlying assumptions. Systems built for human input, such as from crowd workers, and those designed for automation, like LLMs, often operate under distinct assumptions that can introduce friction into the design process. Therefore, it is crucial to be mindful of these embedded assumptions when integrating the two.

In our experience, we have focused on incorporating human input into infrastructures initially designed for automation. For example, the eight-second timeout imposed by Alexa Skills poses a challenge when tasks require longer processing times, such as involving a human worker in the loop. Alexa's infrastructure is primarily structured for predefined tasks and responses, making it ill-suited for freestyle speech, which is common in human-to-human

conversations. Additionally, longer speeches, typical in human interactions, are incompatible with Alexa's strict turn-based model, where only one party can speak at a time. Users often feel uncertain about when the device stops "listening," causing insecurity and interruptions during extended speech. Moreover, the inability to edit speech after Alexa stops listening forces users to wait for a response before making corrections, which is inefficient. The lack of clear indications of the transcribed text also makes it difficult to backtrack and edit lengthy input. Alexa's reliance on text-to-speech (TTS) for delivering responses can further complicate matters, as not all messages are effectively communicated in a single lengthy response, which risks losing user focus. These challenges are not unique to Alexa but seem common across many smart speakers.

Regarding LLMs, a prevalent concern is the potential replacement of human workers. While LLMs can assist crowd workers and are increasingly integrated into company workflows, their adoption may eventually reduce opportunities for human workers due to time and cost efficiencies. However, the situation is more complex than simply swapping out human workers for LLMs. The two configurations face distinct challenges, and removing the human component is not always a straightforward or effective solution. It is important to recognize that while LLMs are highly capable, they cannot seamlessly replace human workers in every context. There remain fundamental differences between the two, and LLMs often require modification to address specific problems effectively.

11.4.3.2 Being aware is as important as being intelligent

Our second design implication emphasizes that in voice-based systems, the focus should shift from enhancing the "intelligence" of the system to improving user interaction. While LLMs perform well in text-based conversations, the real challenge lies in how they handle voice-based interactions. The broader question of whether LLMs can replace humans goes beyond their ability to hold a conversation in text form; it also depends on their understanding of the conventions, expectations, and norms of human conversations in voice-based modalities. Simply making LLMs more "intelligent" will not address the remaining gaps in user experience.

Voice-based interactions, such as phone calls, introduce complexities that are not present in text-based or controlled environments. The lack of visual feedback (e.g., Echo's light ring) and the presence of background noise and interruptions make these interactions more challenging. Studies conducted in quiet, controlled settings, such as those at ECHOPAL and CHATGPT-IN-ECHO, do not fully capture the realities of how users interact with voice assistants in everyday situations. These challenges underline the

need for innovative interaction designs that account for real-world, voice-only contexts where factors like noise and the absence of visual cues are critical. Improving user experience in these scenarios requires more than just better LLMs; it calls for thoughtful, user-centered interaction design that addresses the unique demands of voice-based communication.

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